REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Mathematics and Physics Practice Questions

Grade 9 Advanced | Module 1 Lesson 2 | Mathematics and Physics

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: $v = d/t$; $\rho = m/V$; A = length × width; 1 km = 1000 m; 1 mm = 10^{-3} m.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 9–12.

P01. Which is an SI base quantity?

A. Speed B. Area
C. Time D. Density

P02. Which prefix means one thousandth?

A. Kilo B. Mega
C. Giga D. Milli

P03. How many significant figures are in 0.00450?

A. Three B. One
C. Two D. Five

P04. Which unit is appropriate for speed in SI?

A. m s B. m/s
C. kg/m D. s/m

P05. Why must both sides of a physical equation have the same dimensions?

A. They must have the same numerical value in every unit system B. All physical quantities have identical dimensions
C. They represent the same kind of physical quantity D. Units can be ignored

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P06. Convert 5 mm to metres.

A. 0.005 m B. 0.01 m
C. 0.0025 m D. -0.005 m

P07. Convert 2.4 km to metres.

A. 4800 m B. 2400 m
C. 1200 m D. -2400 m

P08. A car travels 240 m in 7 s. Calculate its average speed.

A. 68.57 m/s B. 17.14 m/s
C. 34.29 m/s D. -34.29 m/s

P09. A sample has mass 160 g and volume 20 cm^3. Calculate its density in g/cm^3.

A. 16 g/cm^3 B. 4 g/cm^3
C. -8 g/cm^3 D. 8 g/cm^3

P10. A rectangle is 4 cm long and 3 cm wide. Calculate its area.

A. 12 cm^2 B. 24 cm^2
C. 6 cm^2 D. -12 cm^2

P11. What is dimensional analysis?

P12. Distinguish a base quantity from a derived quantity.

P13. Explain the meaning of the prefix kilo.

P14. Explain why significant figures are important.

P15. State the rounding rule for multiplying measured quantities.

P16. Convert 7.5 mm to metres.

P17. Convert 10 mm to metres.

P18. Convert 3.6 km to metres.

P19. Convert 4.8 km to metres.

P20. A car travels 360 m in 8 s. Calculate its average speed.

P21. A car travels 480 m in 9 s. Calculate its average speed.

P22. A sample has mass 240 g and volume 30 cm^3. Calculate its density in g/cm^3.

P23. A sample has mass 320 g and volume 40 cm^3. Calculate its density in g/cm^3.

P24. A rectangle is 6 cm long and 4 cm wide. Calculate its area.

P25. A rectangle is 8 cm long and 5 cm wide. Calculate its area.

P26. Check whether d = vt is dimensionally consistent when d is distance, v is speed and t is time.

P27. A student writes speed = distance × time. Identify the problem using units.

P28. Calculate 2.4 m × 3.12 m and report the correct significant figures.

P29. Calculate 12.35 cm + 0.4 cm with appropriate precision.

P30. An answer is numerically plausible but has units kg/s for a speed. Can it be accepted? Explain.